

CHE 330 Week 3 Homework

Assigned 9/10/2025, Due 9/16/2025 11:59 PM

1. Compare and contrast 3 methods of polymerization. 

Chain-growth polymerization

Mechanism: initiation to propagation to termination. A reactive center (radical, cation, anion, or coordination center) adds monomer one at a time to a growing chain. 

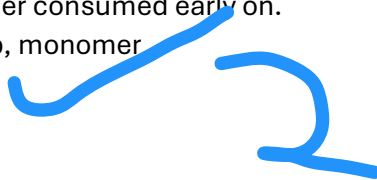
Step-growth polymerization

Mechanism: any two reactive functional groups can react (monomer–monomer, monomer–oligomer, oligomer–oligomer). Chains gradually grow by condensation steps.

Ring-opening polymerization

Mechanism: opening of a cyclic monomer (lactams, lactones, epoxides) to give linear polymer. Can be cationic/anionic/coordination/enzymatic.

2. Highlight key differences between step and chain polymerization.

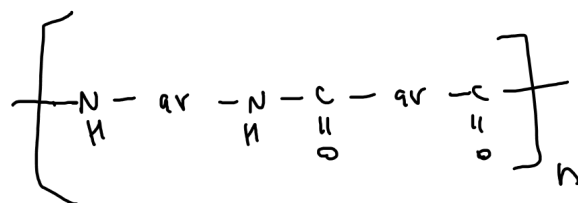
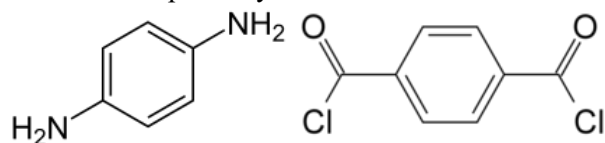
Key differences between the two are what molecules react, speed of mw build up, and amounts of monomers consumed at the start of the reaction. Step growth can have any molecules react, slow mw builds up, and large amounts of monomer consumed early on. Chain growth only active chain ends up reacting, fast mw builds up, monomer consumption decreases with time. 

3. Identify the differences in the polymer structure, starting materials and method of formation of **Nylon 6** and **Nylon 6,6** that are produced commercially.

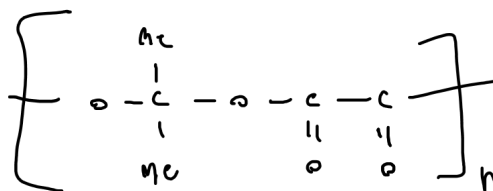
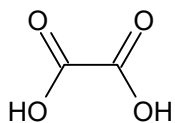
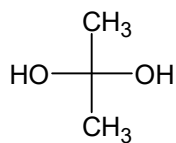
Nylon 6: it's formed from ring opening and its starting material is a single repeating unit of 6 carbons

Nylon 6,6: formed from polycondensation, and its starting material is two different monomers (hexamethylenediamine and adipic acid) 

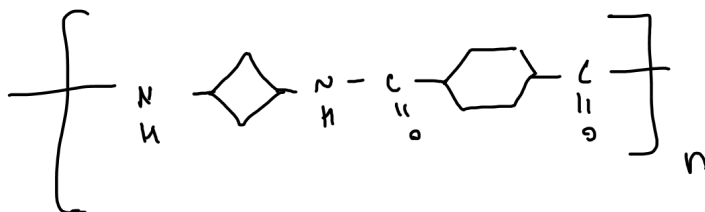
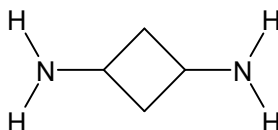
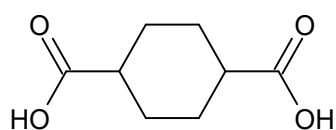
4. For each of the following monomer/starting materials, draw the polymer structure formed
p-Phenylenediamine and Terephthaloyl chloride



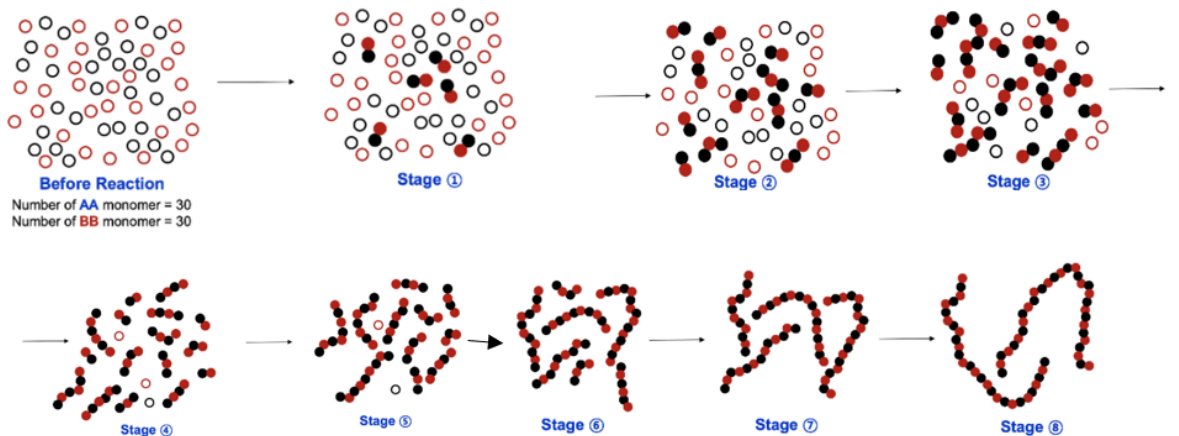
Propane-2,2-diol and oxalic acid



cyclohexane-1,4-dicarboxylic acid and cyclobutane-1,3-diamine



5. Consider the step polymerization of 60 bifunctional molecules shown below.
- Compute the number of functional groups at each stage of the reaction.
 - Determine the extent of the reaction and degree of polymerization at each stage of the reaction.
 - Make a plot of the extent of reaction, p (on the x-axis) vs. degree of polymerization, x_n (on the y-axis), and discuss the meaning of the plot.





χ : 4 bonds formed

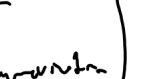
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